1. Overview

In this assessment you will identify a real world problem that can be addressed with biological or biomedical data, and carry out a complete bioinformatics project in the **R programming language**. Your work should be submitted as a GitHub repository. You are free to choose the problem, but it must be realistic, clearly defined, and answerable with publicly available data and the methods covered in the module.

2. Learning Outcomes

On completing this assessment you should be able to:

- Formulate a biological or health-related problem as a data-driven question.
- Source, import, clean and document a real biological dataset.
- Apply appropriate statistical, visualisation and/or machine learning methods in R.
- Interpret results critically and relate them to the real-world problem.
- Present reproducible, version-controlled work on GitHub with clear code, figures and a written report.

3. The Task

1. Choose a real-world problem.

   Examples of areas (not a restricted list):

   - Disease biomarker discovery (e.g. differential gene expression in cancer vs. normal tissue)
   - Classifying patients or samples (e.g. tumour subtype, disease vs. healthy) using gene expression or clinical features
   - Sequence analysis (e.g. comparing viral variants, identifying conserved regions, GC content and motif analysis)
   - Phylogenetics (e.g. evolutionary relationships among pathogens or species)
   - Antimicrobial resistance or pathogen genomics
   - Protein structure/function analysis and annotation
   - Pathway and gene ontology enrichment for a disease of interest
   - Public health or epidemiological genomic data (e.g. outbreak tracking)

    - Agriculture, food or environmental genomics (e.g. crop stress response, microbiome profiling)

2. Obtain a real dataset from a public source.
3. Analyse the data in R, following a sensible workflow: import, quality control and cleaning, exploratory analysis, main analysis, interpretation.
4. Report your findings in a written report with supporting code.

4. Suggested Data Sources

- NCBI (GenBank, GEO, SRA, PubMed)
- Ensembl and UCSC Genome Browser
- UniProt and Protein Data Bank (PDB)
- The Cancer Genome Atlas (TCGA) / GDC portal
- Bioconductor experiment data packages (e.g. airway, golubEsets, ALL)
- Kaggle and UCI Machine Learning Repository (biological/medical datasets)
- Other reputable public repositories (cite the source and licence)

5. Technical Requirements

- All analysis must be done in **R** (RStudio recommended). Use of **R Markdown** is strongly encouraged.
- Use appropriate packages, for example tidyverse, ggplot2, Biostrings, limma, DESeq2, edgeR, clusterProfiler, ape, phangorn, caret, e1071, pheatmap.
- Your project must include at least:
    - Data import and cleaning/preprocessing
    - Exploratory data analysis with meaningful visualisations
    - One or more appropriate analytical methods (statistical testing, clustering, differential analysis, sequence analysis, or a predictive model such as regression or SVM)
    - Interpretation of results in the context of the problem
- Code must be well commented and run from start to finish without errors

6. Deliverables

1. A GitHub repository (public) containing all project work.
2. README.md at the repository root: your written report (approximately [1,500–2,500] words, excluding references), with figures embedded.
3. R code (.R, .Rmd or .qmd) in a code/ folder that reproduces all results and figures.
4. Rendered output (HTML/PDF) of any R Markdown/Quarto notebooks, committed to the repository.
5. Data: small datasets in a data/ folder, or a script that downloads them.

7. Suggested README (Report) Structure

1. Title and abstract (max. 150 words)
2. Introduction: the real-world problem, why it matters, and your aim/hypothesis
3. Data: source, description, size, variables, and any ethical or licensing notes
4. Methods: workflow and justification of the techniques and R packages used
5. Results: figures, tables and statistics with clear captions
6. Discussion: interpretation, limitations, and real-world implications
7. Conclusion and possible future work
8. Reproducibility: how to run the code, required packages and versions
9. References (consistent citation style, including datasets and packages)

8. Submission Instructions

- Submit the **URL of your GitHub repository** through LMS before **15.11.2026**.
- Name the repository with your student registration number.
- The last commit before the deadline is the one assessed. Commits made after it are ignored.
- Commit regularly (at least [5] commits) with meaningful messages; the history shows that the work is your own.